

Lab-4

Software Requirement Specification (SRS)

Student Placement Portal
Software Engineering — Lab Record

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1. Aim

To prepare a complete Software Requirement Specification (SRS) document for the selected software project, the Student Placement Portal, covering the overall description of the product, its functional and non-functional requirements, external interfaces, design constraints, and other applicable requirements.

2. Introduction

2.1 Purpose

This document specifies the software requirements for the Student Placement Portal, a web-based platform intended to support college students during campus placement preparation and recruitment drives. It states what the system is expected to do, the conditions under which it must operate, and the quality attributes it must satisfy.

The intended readers of this document are the project team members, the faculty guide evaluating the lab record, and any developer who later takes up maintenance of the system. The requirements listed here are treated as the agreed baseline for the design and implementation phases that follow.

2.2 Scope

The Student Placement Portal brings placement-related information into a single place. A student who logs in can browse the companies that visit the campus, read their eligibility criteria and CTC details, see how many rounds the recruitment process has, and go through interview experiences written by seniors who were placed in those companies. The portal also carries placement reports for past seasons.

In addition to information, the portal provides a Mock Online Assessment (OA) module. A student can attempt a set of placement-oriented questions, submit the attempt, and receive a score along with feedback on the answers.

Content on the portal is managed through an Admin Control Center. Company records, placement reports, and mock question banks are added and edited by the administrator. Placement experiences submitted by placed students are held in a pending state and become visible to other students only after the administrator verifies and approves them.

The following items are outside the scope of this version of the system:

- Online application or registration for a live placement drive. The portal is an information and preparation tool, not an application-submission system.

- Direct login or dashboard access for recruiters and company HR.
- Integration with external job portals such as LinkedIn, Naukri or company career pages.
- Automatic import of academic records from the institution ERP. Eligibility values are entered as part of the company record.
- Any guarantee of placement. The portal only supports preparation and access to information.

2.3 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirement Specification
OA	Online Assessment, the first screening test conducted by most recruiters
CTC	Cost to Company, the total annual package offered to a candidate
TPO	Training and Placement Officer
Admin	Administrator who manages portal content through the Admin Control Center
Experience	A write-up submitted by a placed student describing the recruitment process of a company
FR	Functional Requirement
NFR	Non-Functional Requirement
RBAC	Role Based Access Control
CRUD	Create, Read, Update and Delete operations on a record

Table 1: Definitions and abbreviations used in this document

2.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- Lab-3 record of this project: Requirements Engineering and Elicitation, Student Placement System. The stakeholder list, questionnaire results and user stories in that record are the input to this SRS.
- R. S. Pressman, Software Engineering: A Practitioner's Approach, chapters on requirements modelling.
- Project repository maintained on GitHub, as set up in Lab-1.

2.5 Overview of the Document

Section 3 gives the overall description of the product, its users, and the environment it runs in. Section 4 lists the functional requirements module by module. Section 5 covers the external interfaces. Section 6 lists the non-functional requirements. Sections 7 and 8 cover the remaining requirements, constraints and traceability back to the elicitation work done in Lab-3.

3. Overall Description

3.1 Product Perspective

The Student Placement Portal is a new, self-contained product. It is not a replacement for or an extension of any existing software used by the college, and it does not depend on any external system to function.

All the data it uses, including company records, eligibility values, placement reports, mock questions and experiences, is maintained inside the portal itself.

The system follows a three-tier structure. The browser-based front end handles presentation, an application server handles request processing, authentication and evaluation of mock tests, and a relational database stores all persistent data. Students and administrators use the same application but see different interfaces depending on their role.

3.2 Product Functions

At a high level the portal provides the following groups of functions:

1. Account creation and role-based login for students and administrators.
2. Browsing and searching the company directory, with details of role, CTC, eligibility criteria and the number of recruitment rounds.
3. Viewing placement reports for previous recruitment seasons.
4. Reading verified interview and placement experiences, filtered by company.
5. Attempting mock online assessments and receiving a score with per-question feedback.
6. Submitting a placement experience for review, available to students who have been placed.
7. Administrative management of companies, reports, mock questions and submitted experiences.

3.3 User Classes and Characteristics

User Class	Description	Technical Skill	Frequency of Use
Student (general)	Any enrolled student preparing for placements. Browses companies, reads experiences and attempts mock tests.	Basic. Comfortable with web applications on a phone or laptop.	Daily during the placement season
Placed Student	A student who has received an offer. Has all student privileges and can additionally submit a placement experience.	Basic	Occasional
Administrator	Member of the placement cell or a designated coordinator. Maintains all portal content and approves submitted experiences.	Moderate. Familiar with content management and report handling.	Weekly, and daily during active drives

Table 2: User classes of the Student Placement Portal

3.4 Operating Environment

Component	Requirement
Client side	Any modern browser (Chrome, Firefox, Edge, Safari) on a desktop, laptop or Android/iOS phone. No installation required.
Screen sizes	The interface is responsive and is expected to work from a 360 px wide phone screen upwards.
Server side	Application server running on a Linux host, deployed either on a college server or on a cloud instance.
Database	A relational database such as PostgreSQL or MySQL.
Network	An active internet connection is required. The portal has no offline mode.

Component	Requirement
Version control	Source code is maintained in a GitHub repository, with the main branch kept deployable at all times.

Table 3: Operating environment

3.5 Design and Implementation Constraints

- The system must be delivered as a web application. Native mobile applications are not part of this phase.
- Only college-issued email addresses are accepted at registration, so that access stays limited to enrolled students.
- Passwords must be stored in hashed form. Plain-text storage of any credential is not permitted.
- Experience write-ups submitted by students must not be published directly. An approval step by an administrator is mandatory.
- The project is developed within a single academic semester by a team of four, which limits the feature set to the modules listed in Section 4.
- All code changes must go through the GitHub repository, using feature branches and pull requests rather than direct commits to the main branch.
- The portal is scoped to one institution per deployment. Multi-college hosting is not supported.

3.6 Assumptions and Dependencies

- Students have valid institutional email credentials issued by the college.
- The placement cell supplies accurate company data, eligibility values and placement reports for entry into the portal.
- Experiences submitted by placed students are written in good faith. Accuracy is checked by the administrator during review.
- The mock OA question bank is prepared and maintained by the administrator, and is not generated automatically.
- Hosting and a domain are made available by the institution for deployment.
- If the eligibility rules of the college change during a season, the corresponding company records are updated by the administrator.

4. Functional Requirements

The functional requirements are grouped by module. Each requirement carries an identifier that is used later in the traceability table in Section 8. Requirements are written so that each one can be tested independently.

4.1 User Registration and Authentication

ID	Requirement	Priority
FR-1	The system shall allow a student to register using a college email address, name, branch, batch year and password.	High

ID	Requirement	Priority
FR-2	The system shall reject registration if the email address is already in use or does not belong to the institutional domain.	High
FR-3	The system shall authenticate a registered user with an email and password combination and start a session on success.	High
FR-4	The system shall lock a user out for a defined interval after five consecutive failed login attempts.	Medium
FR-5	The system shall allow a user to reset a forgotten password through a link sent to the registered email address.	Medium
FR-6	The system shall identify the role of the logged-in user as student or administrator and load the corresponding interface.	High
FR-7	The system shall allow a user to log out, after which the session is invalidated.	High

4.2 Student Profile

ID	Requirement	Priority
FR-8	The system shall allow a student to view and edit profile details including name, branch, batch, CGPA and number of active backlogs.	High
FR-9	The system shall allow a student to mark placement status as placed, and to record the company name and role when doing so.	Medium
FR-10	The system shall use the CGPA and backlog values in the profile to indicate, on each company page, whether the student meets the listed criteria.	High
FR-11	The system shall maintain a record of every mock test attempted by the student, with the date and the score obtained.	Medium

4.3 Company Information Module

ID	Requirement	Priority
FR-12	The system shall display a directory of companies that recruit on campus, showing the company name, role offered and CTC.	High
FR-13	The system shall allow a student to open a company page containing the full job description, eligibility criteria, CTC breakup, service agreement details if any, and the list of recruitment rounds in order.	High
FR-14	The system shall allow a student to search for a company by name and to filter the directory by role, minimum CGPA and CTC range.	High
FR-15	The system shall display, against each company, the count of verified experiences available for it.	Medium
FR-16	The system shall allow a student to bookmark a company so that it appears in a shortlist on the dashboard.	Low

4.4 Placement Reports

ID	Requirement	Priority
FR-17	The system shall display placement reports for past seasons, showing the number of students placed, the number of visiting companies, and the highest and average package for the batch.	High
FR-18	The system shall allow a student to view a report branch-wise and year-wise.	Medium

ID	Requirement	Priority
FR-19	The system shall allow a student to download a placement report as a PDF file.	Low

4.5 Placement Experience Module

ID	Requirement	Priority
FR-20	The system shall allow a student whose placement status is marked as placed to submit an experience describing the company, the role, each round faced, the questions asked and the final outcome.	High
FR-21	The system shall store every submitted experience with the status Pending and shall not display it to other students while it is in that state.	High
FR-22	The system shall allow a student to view the current status of the experiences they have submitted, as Pending, Approved or Rejected.	Medium
FR-23	The system shall display approved experiences on the corresponding company page, with the author name and year of placement.	High
FR-24	The system shall allow a student to submit an experience anonymously, in which case the author name is hidden from other students but remains visible to the administrator.	Low

4.6 Mock Online Assessment Module

ID	Requirement	Priority
FR-25	The system shall allow a student to start a mock test by selecting a topic or a company-specific question set.	High
FR-26	The system shall present questions one at a time along with a countdown timer for the whole test.	High
FR-27	The system shall allow a student to move between questions, change a selected answer, and mark a question for review before submission.	Medium
FR-28	The system shall submit the test automatically when the timer reaches zero.	High
FR-29	The system shall evaluate the submitted test and display the score, the number of correct and incorrect answers, and the time taken.	High
FR-30	The system shall display, for each question, the correct answer and a short explanation after the test is submitted.	High
FR-31	The system shall record the attempt in the student's history so that progress across attempts can be compared.	Medium

4.7 Admin Control Center

ID	Requirement	Priority
FR-32	The system shall allow the administrator to create, update and delete company records along with their eligibility criteria and recruitment rounds.	High
FR-33	The system shall allow the administrator to add and update placement reports for a given batch and academic year.	High
FR-34	The system shall allow the administrator to add, edit and delete questions in the mock OA question bank, including options, the correct answer and an explanation.	High
FR-35	The system shall present all pending experience submissions in a review queue.	High

ID	Requirement	Priority
FR-36	The system shall allow the administrator to approve, edit or reject a submitted experience, and to record a reason when rejecting.	High
FR-37	The system shall allow the administrator to view the list of registered users and to deactivate an account.	Medium
FR-38	The system shall record an audit entry for every content change made through the Admin Control Center, storing the action, the user and the timestamp.	Medium

4.8 Notifications and Announcements

ID	Requirement	Priority
FR-39	The system shall notify a student when an experience they submitted is approved or rejected.	Medium
FR-40	The system shall allow the administrator to publish an announcement that is displayed on the student dashboard.	Medium

Tables 4 to 11: Functional requirements, grouped by module

5. External Interface Requirements

5.1 User Interfaces

The portal is used entirely through a browser. The interface is divided into two views, one for students and one for the administrator, and a user never sees controls that belong to the other role.

Screen	Contents
Login and Registration	Email and password fields, a link to register, and a forgot-password link.
Student Dashboard	Announcements, bookmarked companies, recent mock test scores and quick links to the company directory.
Company Directory	A searchable and filterable list of company cards showing name, role and CTC.
Company Detail Page	Full company description, eligibility criteria, CTC breakup, round-wise process and the list of approved experiences.
Placement Reports	Year-wise and branch-wise report tables with a download option.
Mock Test Screen	One question at a time, a question palette, a countdown timer and a submit button.
Result Screen	Score summary followed by a question-wise review with correct answers and explanations.
Submit Experience	A form capturing company, role, round-wise description, questions asked and outcome.
Admin Control Center	Tabs for companies, reports, question bank, the experience review queue and user management.

Table 12: Main screens of the portal

Every form validates input on the client side before submission, and error messages are displayed next to the field that caused the error. Destructive actions in the Admin Control Center require a confirmation step.

5.2 Hardware Interfaces

The system does not interface with any special-purpose hardware. It requires a standard client device capable of running a browser, and a server with adequate storage for the database and for uploaded report files. No printer, scanner or biometric device is used.

5.3 Software Interfaces

Interface	Description
Browser	The front end communicates with the application server over HTTPS using JSON request and response bodies.
Database	The application server connects to the relational database through a standard driver. All persistent data, including users, companies, reports, questions, attempts and experiences, is stored there.
Email service	An SMTP-based mail service is used to send password reset links and status notifications.
File storage	Placement report files uploaded by the administrator are stored on the server file system, with only the path kept in the database.
GitHub	The source code repository is hosted on GitHub. It is a development-time interface and is not accessed by the running application.

Table 13: Software interfaces

5.4 Communication Interfaces

- All client-server communication takes place over HTTPS on port 443.
- Session state is maintained through a token issued at login and sent with each subsequent request.
- Outgoing email uses SMTP over a secure connection.
- No third-party API is called by the system in this phase.

6. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Usability	A first-time student shall be able to complete registration and open a company page within 5 minutes without any external help or training.
NFR-2	Usability	The layout shall adapt to phone, tablet and desktop widths, since most students access the portal on a phone.
NFR-3	Performance	The company directory shall load within 3 seconds under normal load on a standard broadband connection.
NFR-4	Performance	A submitted mock test shall be evaluated and its result displayed within 2 seconds.
NFR-5	Security	Passwords shall be stored using a one-way hashing algorithm with a salt. They shall never be stored or transmitted in plain text.
NFR-6	Security	Role-based access control shall be enforced on the server side so that a student cannot reach any administrative function by editing the URL.
NFR-7	Security	All input shall be validated on the server to prevent SQL injection and cross-site scripting through experience write-ups.

ID	Category	Requirement
NFR-8	Reliability	Mock test answers shall not be lost if the browser is refreshed during an ongoing attempt.
NFR-9	Availability	The portal shall remain available at least 99 percent of the time during the active placement season, excluding announced maintenance.
NFR-10	Scalability	The system shall support at least 300 concurrent users, which is the expected peak when a mock test is scheduled for a batch.
NFR-11	Maintainability	The code shall be organised into separate modules for authentication, companies, experiences, mock tests and administration, so that a change in one module does not require changes in the others.
NFR-12	Portability	The application shall run on any Linux host with the required runtime and database, without changes to the source code.
NFR-13	Auditability	Every approval, rejection or content change performed by an administrator shall be logged with the user identifier and timestamp.
NFR-14	Data Integrity	A company record that has approved experiences linked to it shall not be deleted outright. It shall be archived instead.

Table 14: Non-functional requirements

7. Other Requirements

7.1 Database Requirements

The database is expected to hold the following main entities, along with the relationships noted against each.

Entity	Key Attributes	Relationship
User	user_id, name, email, password_hash, role, branch, batch, cgpa, backlogs, placement_status	One user submits many experiences and many test attempts
Company	company_id, name, role, ctc, eligibility_cgpa, max_backlogs, branches_allowed, rounds, description	One company has many experiences
Experience	exp_id, user_id, company_id, content, status, is_anonymous, submitted_on, reviewed_by	Belongs to one user and one company
PlacementReport	report_id, batch_year, branch, students_placed, companies_visited, highest_ctc, average_ctc, file_path	Independent record maintained by the administrator
Question	question_id, topic, company_id, text, options, correct_option, explanation	Belongs to a topic or a company-specific set
TestAttempt	attempt_id, user_id, score, total_questions, started_at, submitted_at	Belongs to one user, holds many answers
AuditLog	log_id, user_id, action, entity, timestamp	Written on every administrative change

Table 15: Main database entities

7.2 Version Control Requirements

- The complete source code, along with this SRS and the earlier lab records, is maintained in a single GitHub repository created in Lab-1.

- Each module is developed on a separate feature branch and merged into the main branch through a pull request reviewed by at least one other team member.
- Commit messages state what changed and why, so that the history remains readable during evaluation.
- The main branch is expected to stay in a working, deployable state at all times.

7.3 Legal and Privacy Requirements

- Student CGPA, backlog details and placement status are visible only to the student concerned and to the administrator. They are never shown on public pages.
- Experiences must not contain questions or material that a company has marked confidential. The administrator checks this during review.
- A student may request removal of a submitted experience, and the administrator is expected to act on the request.

8. Traceability with Lab-3

The elicitation work carried out in Lab-3 produced 25 user stories. The table below maps the major findings from that exercise to the requirements defined in this document, so that no elicited need is left unaddressed.

Finding from Elicitation	Addressed By
Students could not confirm eligibility for a company before applying (63 percent reported difficulty).	FR-10, FR-13, FR-14
Placement information was scattered across notice boards, WhatsApp groups and email.	FR-12, FR-13, FR-40
No single place to read what the recruitment process of a company actually looks like.	FR-13, FR-20, FR-23
Students wanted structured preparation material rather than random practice sets.	FR-25 to FR-31
Placement statistics were compiled manually at the end of each season.	FR-17, FR-18, FR-33
Content circulated informally could not be trusted or verified.	FR-21, FR-35, FR-36
Administrative actions on records left no trace.	FR-38, NFR-13

Table 16: Traceability between elicitation findings and requirements

9. Conclusion

This SRS defines the Student Placement Portal in terms of 40 functional requirements spread across eight modules, 14 non-functional requirements, the external interfaces the system exposes, and the constraints under which it will be built. The requirements follow directly from the elicitation carried out in Lab-3, and the traceability table in Section 8 shows that each major finding from that exercise has a corresponding requirement here.

The document is written so that each requirement can be verified independently during testing. It serves as the baseline for the design phase, where the module structure, database schema and interface layouts described here will be worked out in detail.